

NeuroGraphMamba: A Spatial-Temporal Graph State-Space Architecture for Patient-Independent Epileptic Seizure Detection

Ikram Aissioui^{1*}, Naima Boukhiar¹, Sam Guessoum^{2†}, Riad Fellah^{1†}

¹Department of Computer Science, University of Algiers 1, Algiers, Algeria

²Department of Computer Science, Worcester Polytechnic Institute, Worcester, USA

*Correspondence: aissiouikram47@gmail.com

†These authors contributed equally to this work

Purpose: Epileptic seizure detection in previously unseen patients remains one of the most clinically critical and technically challenging problems in medical AI. Despite advances in deep learning, most architectures fail under cross-patient evaluation due to high inter-patient variability in seizure morphology and EEG recording characteristics. Current graph neural network (GNN) approaches encode only spatial electrode relationships, neglecting the directed causal dynamics between EEG frequency bands that are neurophysiologically central to seizure generation. We present NeuroGraphMamba v5, a unified architecture for patient-independent seizure detection that addresses these gaps through three tightly integrated innovations: a causal inter-band graph, a Mamba state-space temporal layer with graph-guided scan ordering, and fully unsupervised test-time adaptation.

Materials and Methods: All experiments were conducted on the CHB-MIT Scalp EEG Database (24 paediatric patients, 18-channel bipolar montage, 256 Hz) and, for cross-dataset evaluation, the Siena Scalp EEG Database (13 adult patients, 29-channel monopolar montage). EEG windows (2 s) were decomposed into five sub-bands (δ , θ , α , β , γ) via five-level discrete wavelet transform (DWT, Daubechies-4) and encoded by a shared per-channel 1D CNN. Two graph structures encode complementary relational information: (i) a spatial adjacency graph (A_{sp}) from electrode proximity and Welch coherence via k-NN ($k=4$); and (ii) a directed inter-band causal graph ($A_{k\theta}$, GrangerEdgeNet) from Granger causality F-statistics (lag $p=3$), cross-band coherence, and a neuroanatomical frequency-ordering prior. Dual GCN layers perform spatial message passing, followed by a causal-graph-guided cross-attention module that sets the Mamba scan order via electrode indegree ranking. Per-patient LoRA ($r=4$) adapters are fine-tuned at inference via a three-stage unsupervised TTA pipeline: (1) batch-norm re-estimation, (2) entropy-minimisation LoRA fine-tuning, and (3) Otsu threshold calibration, without any target-patient labels. Training used focal loss ($\gamma=1.5$), AdamW, and cosine annealing. Both datasets were evaluated under strict leave-one-patient-out (LOPO) cross-validation with 95% bootstrap confidence intervals ($n=500$); the model was applied to Siena without retraining.

Results: NeuroGraphMamba v5 achieved a mean AUC-ROC of 0.914 [95% CI: 0.876–0.948], mean sensitivity of 85.2% [80.1–90.1], and mean specificity of 88.2% [83.6–92.7] across all 24 patients under strict LOPO cross-validation with zero target-patient labels at any stage. Twenty-one of 24 patients exceeded an AUC of 0.868; three patients (chb11: 0.997, chb03: 0.994, chb22: 0.994) achieved near-perfect discrimination. The GrangerEdgeNet inter-band adjacency independently recovered the neurophysiologically established $\delta \rightarrow \theta$ ($w=1.00$) and $\theta \rightarrow \alpha$ ($w=0.75$) hierarchy and identified $\beta \rightarrow \gamma$ ($w=0.54$) as the dominant high-frequency coupling, consistent with high-frequency oscillation involvement in ictal propagation. Midline and fronto-central electrodes (CZ-PZ, FP2-F4, FZ-CZ) emerged as the highest-indegree nodes, aligning with known paediatric seizure propagation pathways. Model calibration yielded an Expected Calibration Error (ECE) of 0.154. Applied without retraining to Siena, NeuroGraphMamba achieved an AUC-ROC of 0.877, sensitivity of 83.5%, and specificity of 81.2%, demonstrating transfer across montage types and age groups.

Conclusions: NeuroGraphMamba v5 demonstrates that integrating directed causal inter-band graph reasoning with Mamba state-space temporal modelling and fully unsupervised test-time adaptation yields robust patient-independent EEG seizure detection without any labelled data from the target patient. The architecture's causal graph outputs are neurophysiologically interpretable, independently recovering established frequency coupling hierarchies and addressing a primary barrier to clinical deployment.

Medical/clinical significance: Automated patient-independent seizure detection can reduce the clinical burden of continuous EEG monitoring in ICUs and ambulatory settings, where expert visual review is both time-intensive and error-prone. By eliminating the need for labelled calibration recordings from each new patient, NeuroGraphMamba directly lowers the barrier to deployment in resource-constrained environments and supports timely clinical decision-making in epilepsy management.

Prospect of application: NeuroGraphMamba targets deployment as a real-time seizure monitoring module in hospital ICU and ambulatory EEG telemetry units, operating on standard 18-channel scalp recordings without patient-specific labelling, supporting clinicians in timely seizure identification, alert generation, and clinical decision support.

References

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